Sinan Eyüboğlu

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EXPERIENCE

Turkish-German University | *Undergraduate Researcher* *Istanbul, Turkey* | **August 2023 - 2025**

- Conducted bioinformatics-driven genomic analysis, integrating large-scale datasets from TCGA and GEO
- Developed models (SVM, RandomForest, LR) for cancer classification and biomarker identification
- Applied statistical techniques for gene expression and survival analysis, analyzing PPI networks
- Explored genomic predictors of drug response across multiple cancer cohorts
- Applied deep learning approaches (GCN, LSTM, GRU) to model intrinsically disordered protein dynamics
- Strengthened expertise in computational biology, data visualization, and interdisciplinary research

Gebze Technical University | *Bioinformatics Intern* *Gebze, Kocaeli* | **July 2024**

- Conducted transcriptome and proteome analyses on Alzheimer's Disease datasets
- Performed differential expression analysis and gene correlation studies
- Focused on omics data integration and co-expression network analysis

Vision Biotechnology | *Laboratory Intern, R&D* *Istanbul, Turkey* | **January 2025**

- Engaged in R&D for diagnostic test kits, focusing on lateral flow assays and PCR-based diagnostics
- Learned membrane immobilization and conjugation buffer preparation for biomarker detection
- Conducted quality control and assay validation for lateral flow and qPCR tests
- Gained expertise in molecular biology techniques, including nucleic acid amplification and fluorescence detection

EDUCATION

Turkish-German University | *Bachelor of Science in Molecular Biotechnology* *Beykoz, Istanbul* | **2021 - 2025**

- Coursework includes bioinformatics, molecular biology, and biotechnology techniques, both in silico and in the wet lab
- Lab techniques include PCR, SDS-PAGE, cloning, chromatography, etc., providing a solid foundation in experimental biotechnology
- Active participation in bioinformatics and omics research projects, combining computational and laboratory skills

Gebze Technical University | *Master of Science in Artificial Intelligence* *Gebze, Kocaeli* | **2025 - July 2027**

- Advanced coursework in core AI disciplines, including machine learning, deep learning and applications, and pattern recognition
- Focus on data-driven methodologies through courses like statistical data analysis and big data analytics
- Thesis: Computational Modeling of Post-Transcriptional Regulatory Heterogeneity in Alzheimer's Disease.

PROJECT EXPERIENCE

PDAC Biomarker Discovery and Machine Learning Classification *Turkish-German University* | **2023**

- Developed and tested machine learning models to classify PDAC samples
- Performed feature selection and analysis of hub genes for early diagnosis biomarkers
- Performed survival analyses to evaluate the clinical significance of the biomarkers using the developed machine learning models.

Unraveling Genetic Interactions in Alzheimer's Disease *Gebze Technical University* | **2024**

- Conducted large-scale RNA-seq data analysis on AD patient samples
- Focused on the correlation of APOE genotypes with gene expression
- Built gene co-expression networks and identified key pathways involved in disease progression

Genomic Predictors of Drug Response Across Cancer Cohorts *Turkish-German University* | **2025**

- Investigated how genomic alterations such as copy number variation, mutations, and gene expression influence patient response to specific anticancer drugs across multiple TCGA cancer cohorts
- Focused on Paclitaxel in BRCA, Gemcitabine in PAAD, and 5-FU in STAD cohorts, using sample-level sensitivity/resistance information
- Performed survival modeling using Cox regression, extracting hazard ratios and C-indices to quantify effects.

Predicting the Temporal Dynamics of IDPs Using DL *Turkish-German University* | **2025**

- Conducted as a Bachelor's thesis, focusing on the conformational dynamics of IDPs
- Investigating disordered protein dynamics using deep learning and MD simulations
- Developed GCN & GRU models to predict protein conformational changes
- Compared force field accuracy in structural behavior analysis

Predicting Cryptic Druggable Pockets in IDPs Using DL *Turkish-German University* | **2026**

- Developed a novel deep learning pipeline combining GNNs and LSTM networks to predict the formation of transient, "cryptic" druggable pockets from MD simulations.
- Designed integrative feature sets combining dynamic, geometric, and chemical descriptors for protein atoms
- Aimed to identify therapeutic targets in "undruggable" proteins such as amyloid-beta, relevant to neurodegenerative disorders

RegSpliceNet: Latent Modeling of Splicing Regulation in Alzheimer's Disease Gebze Technical University | **2026**

- Developed a probabilistic encoder-decoder (VAE-style) model in PyTorch that maps each donor's exon-inclusion/skipping read counts to a 50-dimensional latent summary
- Applied the frozen model to new donors and external cohorts, and tested latent scores against Braak, CERAD and cognitive phenotypes with covariate adjustment

Multi-omics Integration, AI, and Organoid Technologies for Investigating Cancer Drug Efficacy and Resistance
Dokuz Eylül University | Incoming Researcher (BAP Scholarship)

- Funded by Dokuz Eylül University Scientific Research Projects (BAP), Project No. 4109
- Principal Investigator: Prof. Dr. Şerife Esra Erdal Bağrıyanık

PUBLICATIONS

1. **Eyuboglu, S., Uversky, V. N., Alpsoy, S. & Coskuner-Weber, O.** (2024). Key genes and pathways in the molecular landscape of pancreatic ductal adenocarcinoma: A bioinformatics and machine learning study. *Computational Biology and Chemistry, 113*, 108268. https://doi.org/10.1016/j.compbiolchem.2024.108268
2. **Coskuner-Weber, O., Akkum, F. I., Caglayan, S. I., Akbayrak, I. Y., Uversky, V. N., & Eyuboglu, S.** (2026). How do AI ensemble pipelines treat disorder? A head-to-head comparison on α-synuclein. *Journal of Biomolecular Structure and Dynamics*, 1–18. https://doi.org/10.1080/07391102.2026.2630328
3. **Coskuner-Weber, O., Eyuboglu, S., & Uversky, V. N.** (2026). Combining AI, simulation, and experiment to map protein folding. In *Comprehensive Biophysics* (2nd ed.). Elsevier. https://doi.org/10.1016/B978-0-323-99507-8.00081-0
4. **Eyuboglu S., Luleci H. B.,** & **Cakır T.** From Protection to Risk: APOE Genotypes Reshape Gene Co-Expression Networks in Alzheimer's Disease. *Under review.*
5. **Alpsoy S., Eyuboglu S. & Sezerman O.U.** Molecular programs associated with paclitaxel resistance encode prognostic information in breast cancer. *Under review.*
6. **Coskuner-Weber, O., Eyuboglu, S., & Uversky, V. N.** Druggability Beyond Structure: State-Centric Targeting of Intrinsically Disordered Proteins in Neurodegeneration and Cancer. *Under review.*

7. **Eyuboglu, S. (in press).** Multi-domain and Complex Protein Structure Predictions. In *In Silico Protein Structure and Function: From AI-Based to Hybrid Quantum Computing Approaches (1st ed.). Elsevier.*
8. **Eyuboglu, S., & Coskuner-Weber, O. (in press).** Mutagenesis and Protein Structure-Function Predictions. In *In Silico Protein Structure and Function: From AI-Based to Hybrid Quantum Computing Approaches (1st ed.). Elsevier*.

SKILLS

- **Machine Learning & Deep Learning** *PyTorch, PyTorch Geometric, scikit-learn*
- **Programming Languages:** *Python, R, Bash*
- **Bioinformatics Tools:** *DESeq2, ClinVar, Cytoscape, BLAST*
- **Computational Environments & Tools:** *High-Performance Computing, Linux/Unix Command Line*
- **Wet Lab:** *PCR, SDS-PAGE, Cloning, Chromatography*
- **Communication:** *English, German, Turkish*

CERTIFICATES

- Bioconductor for Genomic Data Science
- Command Line Tools for Genomic Data Science
- Reproducible Research
- Exploratory Data Analysis
- Algorithms for DNA Sequencing
- Getting and Cleaning Data
- Python for Genomic Data Science
- Introduction to Genomic Technologies
- R Programming

REFERENCES

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